

Generative AI Foundations for Engineers

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Introduction

Agenda

- Brief introduction to the people in the room
- Structure for today
 - *Brief* lectures
 - Activities
 - Challenges (what is scored!)
- Module 1: Generative AI Foundations & LangChain Basics
- Module 2: Prompt Engineering & Model Customization
- Module 3: Retrieval-Augmented Generation (RAG) & External Data Integration
- Module 4: Agents & Tool Use
- Wrap up, announcement of winners!

A brief survey:

Pollev.com/clairsullivan399

Schedule

9:00 - 9:15	Introduction
9:15 - 9:30	Module 1: Lecture
9:30 - 10:30	Module 1: Hands On
10:30 - 10:40	Module 2: Lecture
10:40 - 12:00	Module 2: Hands On
12:00 - 1:00	LUNCH
1:00 - 1:10	Module 3: Lecture
1:10 - 2:25	Module 3: Hands On
2:25 - 2:35	Module 4: Lecture
2:35 - 3:50	Module 4: Hands On
3:50 - 5:00	Wrap up, awards!

Module 1: Generative AI Foundations & LangChain Basics

What is Generative AI (GenAI)?

- Traditional AI (AKA machine learning) focuses on classification, prediction
- GenAI **creates** new content
- Trained on large datasets using deep learning to learn patterns and generate novel output
- Example of GenAI in action?

How Large Language Models (LLMs) Work

- Transformer models: the backbone of LLMs
 - Captures relationships between words across long texts
 - Key innovation in models like GPT, BERT, etc.
- Text is broken into **tokens**
- LLMs have a maximum number of tokens they can process: **context window**
 - When exceeded, older tokens are forgotten (“sliding window effect”)
 - Varies by model
 - Longer context ≠ perfect memory
- LLMs are asked to do things through their **prompts**
- Limitations of LLMs
 - **Hallucinations**
 - Training data limitations (date, subject matter)
 - Computation cost

Introduction to LangChain

- Framework (Python, JS) for building applications powered by LLMs
- Why use it?
 - Open source
 - Simplifies development
 - Has wrappers for most common LLMs, cloud platforms
 - Manages context
 - Integrates with many data types, APIs
 - Chains, agents, tools easily created
- Cons of using LangChain
 - Debugging issues harder compared to direct API calls
 - Latency
 - In *active* state of development
 - LangGraph migration

Introduction to AWS Bedrock

- Fully managed service that provides foundation models via APIs
 - Anthropic, Meta, Mistral, Amazon Titan
- Enables creation of GenAI applications without managed infrastructure
- Seamless AWS integration with tools like S3, SageMaker, IAM, etc.

Structure for the Hands On Work

- Three topic areas
 - Code generation
 - Structured data analysis
 - Unstructured data analysis
- You don't have to stick with one topic area or submit solutions for every question
- Start by going through the “Activity” notebooks
- You will be scored based on the “Challenges”

Ground Rules

- **Have fun!!!**
- Experiment
- Engage with everyone on your team
- Extra points can be awarded at any time for great *shared* learnings
- Extra points awarded for early submissions
- *The Honor System: many of the challenges can be solved to some extent with traditional programming. For the answers you submit, please be honor-bound to submit your answers created via a GenAI solution.*

Some Notes for Module 1

- Don't worry (too much) about rewriting the prompts
 - Just try to ask good questions through the Human Prompt
- Some problems might be easy to solve with simple coding, but let's stick to making the LLMs do it for us
- All code, slides can be found on the repo

[**https://github.com/ClairSullivan-Associates/genai_foundations**](https://github.com/ClairSullivan-Associates/genai_foundations)

How to get started with AWS Workshop